

Human diet of premodern mainland Japan: a meta-analysis of carbon and nitrogen stable isotope ratios

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Abstract The development of the modern industrialized food production system has resulted in a homogeneous human diet worldwide. However, it is not clear whether a developed food production system led to a homogenized human diet also in ancient societies. Due to the lack of large archaeological datasets, we know little about the chronological trends and ancient circumstances of dietary homogenization. Here we compiled carbon and nitrogen stable isotope ratios, indicators of palaeodiet, of adult human skeletons from premodern mainland Japan (AD 1603–1868, $n = 318$) to investigate chronological changes in diet. Comparison with datasets from Japan in modern, premodern (Edo), and foraging (Jomon) periods showed that the human diet was rapidly homogenized isotopically in modern times. Premodern people in Japan typically obtained dietary proteins from C3 crops and fish, and the establishment of agriculture created a new isotope dietary niche compared with the foraging period. Dominant protein contributions from agricultural C3 crops cultivated with organic fertilizers and/or rice that are grown in paddy fields with denitrification increased premodern human nitrogen isotope ratios without increasing their carbon isotope ratios. Diet differed according to the social status of individuals or the availability of foods, and a unique diet can be seen in people in higher social classes such as the Shogun family. Meta-analysis of stable isotope ratios of archaeological human skeletons enables a comprehensive understanding of human dietary change through time and regional variations.

Key words: diet, Edo, meta-analysis, stable isotopes, *washoku*

Introduction

The food production industry supports a substantial number of human populations today and has resulted in a homogeneous diet in modern industrialized countries (FAO, 2017). After the appearance and establishment of agriculture and domestication in the early Holocene, increased access to stable food sources led to an increase in the human population worldwide (Chamberlain, 2009; Gage and DeWitte, 2009). Modern food production industries and global commerce networks enable access to a wide range of food sources derived from broad geographic ranges for people living in industrialized countries, and the homogenization of consumption patterns typically appears as the

convergence towards a Western-style diet. Such a ‘global supermarket effect’ and subsequent dietary changes characterize the diet of modern industrialized human populations (Nardoto et al., 2006). The appearance of food production in foraging societies and its globalization and industrialization in the Holocene shape contemporary and future human feeding ecology (Diamond, 2002; FAO, 2017).

Dietary homogenization is usually investigated through stable isotope analyses. Stable isotopes are naturally occurring tracers of diet. In ecosystems, the stable isotopes ratios of carbon and nitrogen ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values, respectively) in organisms systematically vary, primarily by the type of photosynthesis of the producers (Smith and Epstein, 1971; O’Leary, 1988) and the trophic relationships (Minagawa and Wada, 1984; Schoeninger and DeNiro, 1984), respectively. Specific categories of food sources (e.g. C3 plants, herbivores, carnivores, or marine organisms) show specific $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values, which can be used to trace the food sources of consumers (Lee-Thorp, 2008; Makarewicz and Sealy, 2015). Stable isotope ratios of proteinaceous tissues, such as bone collagen and hair keratin, mainly reflect those

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of protein fractions of diet and typically represent the protein contribution of food sources (Tieszen and Fagre, 1993; Howland et al., 2003). Bone collagen is the material typically analyzed from archaeological human and faunal skeletons. The turnover rate of bone collagen is relatively small in human adults, and stable isotope ratios of bone collagen represent the average diet of the previous 10–20 years before the death of the individual and are less prone to short-term fluctuations (Hedges et al., 2007).

Studies using carbon and nitrogen stable isotope analysis have revealed a homogeneous diet in the modern industrialized human population (Nardoto et al., 2006; Valenzuela et al., 2012; Kusaka et al., 2016; Lee et al., 2021), but chronological changes from ancient times have been little studied. The $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of modern human hairs and fingernails are fairly homogeneous in industrialized countries, such as the United States, Europe, Japan, and South Korea, due to Westernization and the ‘global supermarket effect,’ although there remain small regional variations (Nardoto et al., 2006; Valenzuela et al., 2012; Kusaka et al., 2016; Lee et al., 2021). However, due to the lack of detailed, comprehensive, and time-series data on $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of ancient human populations, we know little about the chronological trends and prehistorical circumstances of the human diet. For example, has the dietary homogenization seen in modern industrialized countries developed rapidly or gradually? How dominant is the influence posed by the development and establishment of food production systems? Are there regional differences? Much more data on large-scale trends of stable isotope ratios analyzed in archaeological human populations are needed to elucidate the chronological change of dietary homogenization, although a few recent meta-analyses present such evidence, on a limited but representative data set of the world (Bird et al., 2021), or for specific regions, such as the British Isles (Bird et al., 2022).

To investigate chronological changes in dietary homogenization in past human societies, the premodern period of Japan provides one of the most ideal cases. The premodern period, also called the Edo period (AD 1603–1868), in Japan was characterized by the development of food production systems and commerce networks that shaped modern-day traditional food cultures of Japan, known as *washoku* (Ehara et al., 2009; Harada, 2009; Ishige, 2015). Yet, food production and commerce in the premodern period were still practiced at the local level and were neither industrialized nor globalized as can be seen today. Intensive excavation efforts and a number of stable isotope studies allow the chronological comparison of large data sets from Jomon, premodern, and modern-day human populations in Japan. Additionally, the Japan archipelago is located in several different climate zones from subtropic to subarctic, and thereby provides a unique opportunity to compare diets developed in different ecological and cultural contexts. Moreover, the Japan archipelago is isolated from other continents by sea, and the genetic and cultural sequence of the inhabitants is relatively straightforward, especially during the late Holocene (Hanihara, 1991; Hanley, 1997).

The objective of this study was to investigate dietary homogenization in the premodern period of mainland Japan

through the meta-analysis of published $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ ratios of ancient human skeletons. Carbon and nitrogen stable isotope ratios reported in premodern human skeletons were compiled and compared with reported data sets from the foraging period and modern human populations. By comparing the stable isotope datasets obtained from Jomon, premodern, and modern human populations in mainland Japan, we will test the hypothesis that developed food production led to the homogenized human diet even in ancient societies.

Premodern Japan

The Japan archipelago stretches approximately 3500 km from north to south on the northeastern edge of the Pacific Ocean (Figure 1). Regarding archaeological and ecological backgrounds, the archipelago can be divided into three regions consisting of the northernmost part (Hokkaido), the mainland (Honshu, Shikoku, Kyushu, and other small islands), and the southernmost part (Ryukyu or Okinawa). These three regions have different histories of peopling and the development of food production systems (Hanihara, 1991; Ehara et al., 2009; Ishige, 2015; Watanabe et al., 2021). The Ryukyu Islands can be further divided into three subregions, North, Central, and South Ryukyus. During the prehistoric period, North Ryukyu was exclusively under the rein of Tokugawa Shogunate and its culture belongs to mainland Japan rather than Central and South Ryukyus (Pearson, 1969; Asato, 1990, 1991). Therefore, mainland Japan and North Ryukyu are defined as ‘mainland’ for convenience, and data from North Ryukyu are included in this study.

Historical documents provide a wealth of information on the dietary contents of premodern Japan, and the typical diet of premodern people in mainland Japan consisted of rice as the staple food, C3 vegetable crops, and fish as the occasional protein source (Ehara et al., 2009; Harada, 2009; Ishige, 2015). People in mainland Japan during the early Holocene (Jomon period: 16000–2300 BP) subsisted by the foraging of natural resources, such as terrestrial mammals, C3 plants, and marine organisms, and rice agriculture was introduced as a food production system in mainland Japan from the Korean peninsula around 2300 years ago (Habu, 2004). There were introductions of other crops and domesticated animals subsequently, most of the food sources that can be seen today appeared, and modern-day *washoku* culture was shaped in the premodern period (Edo period: AD 1603–1868) (Ehara et al., 2009; Harada, 2009; Ishige, 2015). Eating the meat of quadrupeds was forbidden due to Buddhist precepts, and only chicken and its eggs were sometimes eaten as a protein source of terrestrial animals (Ehara et al., 2009; Harada, 2009; Ishige, 2015). Soybean products, such as soybean curd (*tofu*), were also important protein sources (Hanley, 1997). During the premodern period, the Tokugawa Shogunate ruled the whole of mainland Japan and adopted a policy of seclusion, prohibiting trade with foreign countries except for the Netherlands and China, and allowing what trade there was only via one port (*deijima* in Nagasaki) in Japan (Hanley, 1997; Francks, 2016). Thus, the diet of mainland Japan in the premodern

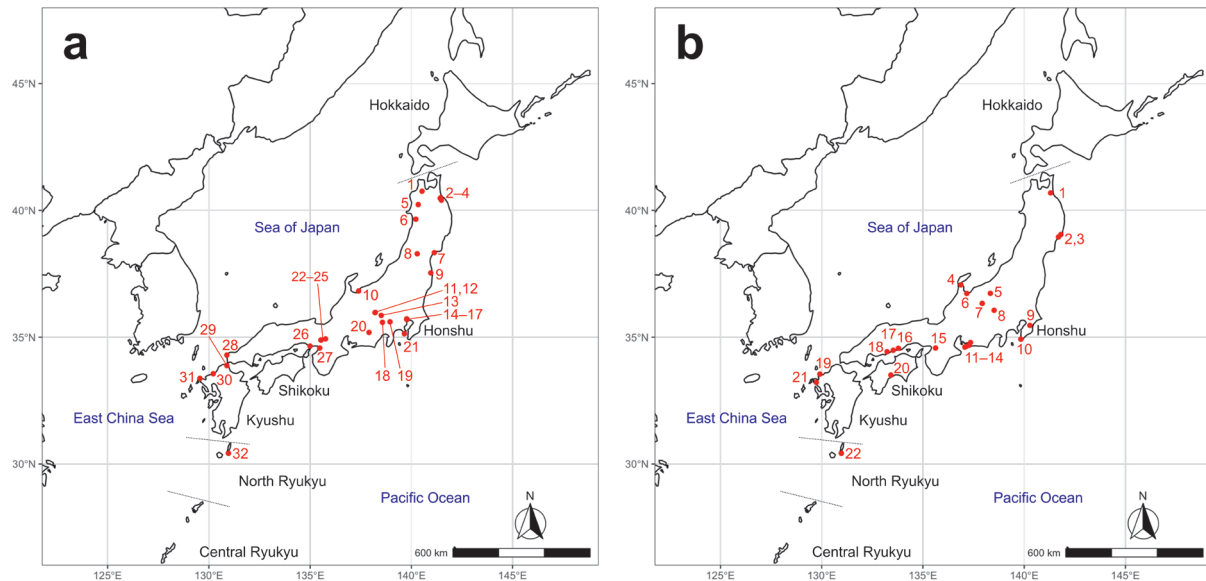


Figure 1. Map showing the location of mainland Japan and the premodern sites used in this study. Honshu, Shikoku, Kyushu, and North Ryukyu are defined as 'mainland' in this study. (a) Premodern site numbering: 1, Kumanashi (8); 2, Sengoku-yashiki 2; 3, Uwano; 4, Araya; 5, Takasu-bouzawa; 6, Kuronumashimotsutsumi-shitatabi-ato; 7, Satohama; 8, Dainichikita; 9, Hachiman-hayashi; 10, Uozukou; 11, Kamiizawa-o'ne; 12, Hodokubo; 13, Shiokawa; 14, Toeizan Kaneiji; 15, Kaneiji Gokokuin; 16, Ikenohata-shichikencho; 17, Hitotsubashi; 18, Anrakuji-higashi; 19, Komekuyama B; 20, Nishiji and Higashiji; 21, Bishamon; 22, Fushimijo-ato; 23, Sendaiji Kurusuyama; 24, Sendaiji West; 25, Sendaiji Hishigatani; 26, Unseiji; 27, Sakai-kango-toshi 871; 28, Doigahama; 29, Kyomachi; 30, Shinohara-higashi I; 31, Miura Anjin no haka; 32, Hirota. (b) Jomon site numbering: 1, Furuyashiki; 2, Nonomae; 3, Nakazawahama; 4, Mibiki; 5, Boji; 6, Otake; 7, Kitamura; 8, Tochiara; 9, Shimo'ota; 10, Sano cave; 11, Inariyama; 12, Yoshigo; 13, Ikawazu; 14, Kawaji; 15, Kou; 16, Funamoto; 17, Tsukumo; 18, Ota; 19, Ohtomo; 20, Itoku; 21, Iwashita cave; 22, Hirota.

period is expected to be relatively homogeneous regardless of the region, and it is thought that it underwent a relatively small chronological changes.

Materials and Methods

Datasets

To compile the stable isotope data reported for premodern human skeletons, a systematic literature search was conducted. Publications in English were searched in Google Scholar with the combinations of the queries such as "stable isotope," "Japan," or "premodern." Publications in Japanese were searched in CiNii provided by the National Institute of Informatics for journal articles, and the Comprehensive Database of Archaeological Site Reports in Japan provided by Nara National Research Institute for Cultural Properties with excavation reports through the combination of queries such as '同位体 (stable isotope),' '近世 (premodern),' or '江戸時代 (Edo period).' Snowball searching was done and literature cited by the obtained publication was also checked. Published datasets on the reported stable isotope ratios of premodern human remains from Japan were also included (Fernandes et al., 2021; Tsutaya and Yoneda, 2021). Only stable isotope data that were reported with numerical values in the table or text, that were measured from bone or tooth collagen, and that can be related to a specific archaeological site were used in this study. Therefore, stable isotope data of hair keratin obtained from historical books (e.g. Minagawa et al., 1986; Maruyama et al., 2018) were not used in this study. Due to the scarcity of available isotope data in premodern

Hokkaido, the northernmost region of Japan, this was excluded from the scope of this study. In addition, due to the possible differences in baseline dietary isotope ratios and the scarcity of faunal isotope data, Central and South Ryukyus, the southernmost regions of Japan, were excluded from the scope of this study.

Comparative datasets of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of modern human hairs from mainland Japan, excluding Hokkaido and Okinawa (Kusaka et al., 2016), human collagen from Jomon periods (Fernandes et al., 2021), human remains from ancient humid C3/C3 regions of the world (Bird et al., 2021), and ancient and modern food sources from Japan (Yoneda et al., 2004) were also referred.

Acceptable entries were extracted from the datasets by applying several criteria for quality control. First, data obtained from non-adult individuals (i.e. <15 years of age) and the dentine portion formed during childhood (e.g. deciduous teeth and the crown portion of M1) were excluded to control for any possible differences resulting from breastfeeding (Tsutaya and Yoneda, 2015) and childhood diet (Tsutaya, 2017). Therefore, the dataset consists of individuals aged older than 15 years. Then, entries with an atomic C/N ratio outside the acceptable range (2.9–3.6 (DeNiro, 1985)) were excluded. Reliable stable isotope ratios of collagen cannot be obtained from such specimens. Finally, any entry lacking either $\delta^{13}\text{C}$ or $\delta^{15}\text{N}$ values was removed. Additionally, one erroneous entry that shows exceptionally low $\delta^{15}\text{N}$ values for human bone collagen (0.9‰: SK17 from Kuronumashimotsutsumi) was excluded, as this was ascribed to misidentification of the taxonomy of the subject specimen. Entries without

Table 1. Summary of the individual stable isotope ratios obtained from different datasets

	$\delta^{13}\text{C}$		$\delta^{15}\text{N}$		No. of individuals	No. of sites	Reference
	Mean	SD	Mean	SD			
Jomon	-17.2	1.8	10.7	2.3	342	19	Fernandes et al., 2021
Premodern	-19.6	1.2	11.2	1.1	318	27	This study
Premodern (conservative)	-19.5	1.2	11.1	1.1	276	20	This study
Modern	-16.5	0.6	10.3	0.6	1035	45	Kusaka et al., 2016
Humid C3/C4	-14.2	4.4	10.3	1.8	922	24	Bird et al., 2021

The results of conservative analysis without individuals with no information on atomic C/N ratio or age class are also shown.

information on atomic C/N ratio or age class were regarded as acceptable to increase the sample size. Even if these entries are excluded from the dataset, the results are, as shown later, essentially the same.

When the dataset of human hairs from modern Japan was used for comparison, corrections for the isotope offset between hair keratin and bone collagen (i.e. +1.4‰ and +0.9‰ for keratin $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values, respectively (O'Connell et al., 2001; Bird et al., 2021)) and the Suess effect (+1.5‰ with modern $\delta^{13}\text{C}$ values (Suess, 1955; Friedli et al., 1986)) were applied. When the dataset of food resources was used for comparison, correction for the isotope offset between food sources and bone collagen (+5.0‰ and +4.0‰ for food $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values, respectively (Lee-Thorp, 2008)) was applied.

Statistical analysis

All statistical analyses were performed with R software, version 4.2.2 (R Core Team, 2022). The 95% and 50% areas of kernel density estimates on the two-dimensional distributions of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values were calculated using the package rKIN, version 0.3.0 (Eckrich et al., 2019). The kernel density estimate performs better irrespective of the sample size and the shape of the distribution of data, compared with other methods such as convex hull and standard ellipse (Eckrich et al., 2019). The level of significance for statistical tests was set as $\alpha < 0.05$. When applying the Mood test for comparison of variance, standardized $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values subtracted with the average were used.

Individual age and chronological period are calculated as the median of the reported ranges. The maximum ranges for the calculation are 15–80 years for age, AD 1600–1926 for the chronological period of the mainland. If the entire range of the individuals' estimated age or chronological period was greater than 25 years or 120 years, respectively, the median was not used for the statistical analyses due to the greater uncertainty. Calibrated age ranges reported in previous literature (Fernandes et al., 2021) were used for individuals with the direct measurement of ^{14}C .

Results

A total of 318 $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values from adult individual human skeletons from the premodern mainland were obtained from published literature after applying several quality controls. The mean $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of skeletal individuals were $-19.6 \pm 1.2\text{‰}$ and $11.2 \pm 1.1\text{‰}$ for the premodern period and $-17.2 \pm 1.8\text{‰}$ and $10.7 \pm 2.3\text{‰}$ for

Jomon period (Table 1, Figure 2). The mean $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of the average of archaeological sites were similar to those of skeletal individuals (Table 2). Even if the total of 42 individuals without information on atomic C/N ratio or age class (adult or non-adult) were excluded as a conservative analysis, the summary statistics were essentially the same (Table 1, Table 2). Therefore, these individuals were included and analyzed as acceptable entries throughout this manuscript. The compiled datasets are available at the IsoArch repository (Salesse et al., 2018; Plomp et al., 2022) with the digital object identifiers 10.48530/isoarch.2021.006 (Tsutaya and Yoneda, 2021) and 10.48530/isoarch.2023.003.

A comparison of the stable isotope ratios of human skeletons with those of possible food sources in ancient and modern Japan (Yoneda et al., 2004) shows that the dietary proteins were typically obtained from a mixture of C3 terrestrial and marine/freshwater food sources during the premodern period (Figure 3). These datasets were compared with the published datasets of stable isotope ratios of human hair from modern Japan ($n = 1035$) (Kusaka et al., 2016), human skeletons from mainland Japan in the Jomon period ($n = 342$) (Fernandes et al., 2021), and archaeological human remains from humid C3/C4 vegetation regions of the world ($n = 922$) (Bird et al., 2021) (Figure 4). A Mann–Whitney *U*-test showed significant differences among Jomon, premodern, and modern individuals for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ (Table 3).

The 50% and 95% probability areas of the two-dimensional $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values calculated by kernel density estimate (Eckrich et al., 2019) were smaller in modern mainland Japan compared with premodern and foraging periods (Table 4, Figure 4). A Mood test showed that variances of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values were significantly smaller in modern Japan compared with premodern and Jomon mainland (Table 5). These results suggest that the diet is isotopically more homogeneous in modern Japan than it was in ancient times. The dataset from premodern mainland Japan has smaller 50% and 95% kernel density areas (Table 4) and significantly smaller variations in $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values (Table 5) compared with the Jomon dataset (Figure 4).

Inter- and intra-population variations in premodern human stable isotope ratios were generally larger than in modern period but smaller than in Jomon period. Considering the populations (i.e. skeletal populations for the archaeological datasets and prefectures for the modern dataset) which have at least three individual data, inter-site variations (i.e. 1SD range of population average $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$)

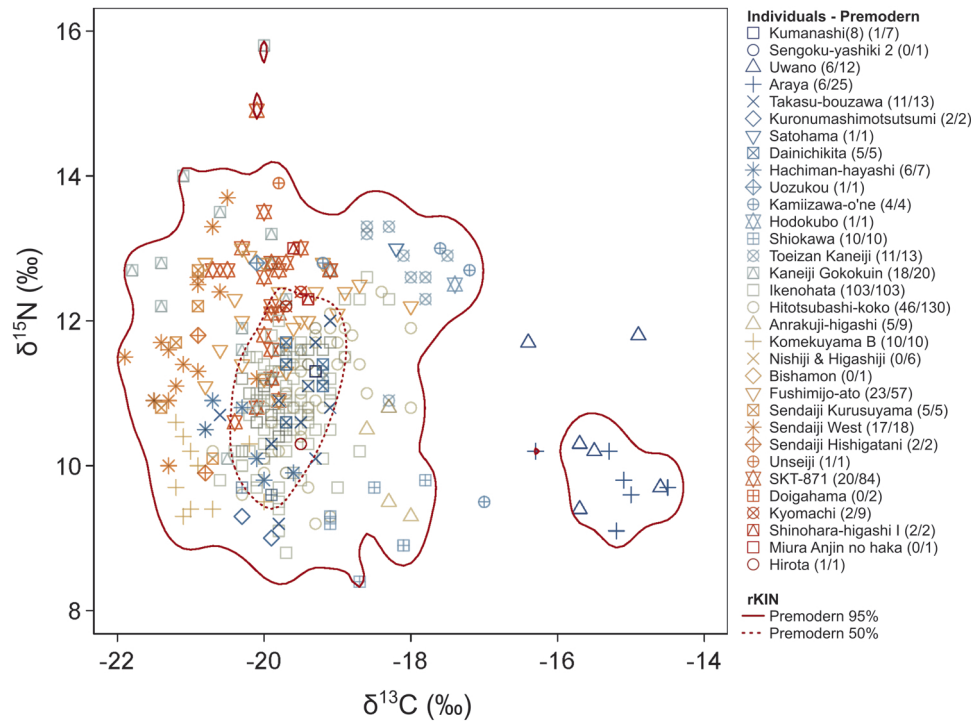


Figure 2. Distribution of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of premodern skeletal individuals from mainland Japan. The 50% and 95% kernel density areas of skeletal values calculated by rKIN (Eckrich et al., 2019) are shown. Numbers in parenthesis are the accepted number and the total number of subject individuals from the site.

Table 2. Summary of the average stable isotope ratios of different sites (Jomon and premodern) or prefecture (modern)

	$\delta^{13}\text{C}$			$\delta^{15}\text{N}$			<i>n</i>
	Mean	SD	Median	Mean	SD	Median	
Jomon	-17.6	1.9	-16.7	10.5	3.0	10.6	21
Premodern	-19.3	1.5	-19.6	11.4	1.2	11.3	27
Premodern (conservative)	-19.3	1.6	-19.7	11.4	1.2	11.4	20
Modern	-16.5	0.2	-16.5	10.3	0.2	10.4	45

The results of conservative analysis without individuals with no information on atomic C/N ratio or age class are also shown.

were greater in the order Jomon, premodern, and modern for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values (Figure 5; see also Table 6). In addition, intra-site variations (i.e. median or distribution of 1SD ranges of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ for individual populations with more than three individual data) were greater in the order Jomon, modern, and premodern for $\delta^{13}\text{C}$ and in the order Jomon, premodern, and modern for $\delta^{15}\text{N}$ (Figure 5). These results show that (i) the variability of average diet among different populations was smaller in more recent periods among Jomon, premodern, and modern periods; and (ii) the variability of diet among individuals in the same population was smaller in more recent periods, except for $\delta^{13}\text{C}$ values in the premodern period.

Relatively small stable isotope differences by individual characteristics (e.g. sex and social class) can be seen. A Mann–Whitney *U*-test showed that male individuals from the premodern mainland showed significantly greater $\delta^{13}\text{C}$ values than females ($U=9201$, $P=0.012$), although the $\delta^{13}\text{C}$ difference is relatively small (i.e. 0.3‰). There was no significant sex difference in premodern human $\delta^{15}\text{N}$

values (Mann–Whitney *U*-test, $U=10636$, $P=0.554$). A Mann–Whitney *U*-test showed that the individuals of the Tokugawa Shogunate family have significantly greater $\delta^{13}\text{C}$ (+1.3‰: $U=3067.5$, $P<0.001$) and $\delta^{15}\text{N}$ (+1.3‰: $U=2796.5$, $P<0.001$) values compared with the other individuals from the premodern mainland Japan, although their stable isotope ratios distribute continuously with the other individuals in the 95% kernel density area (Figure 2). There was no evident trend in the isotope ratios of individuals from the different chronological periods or biological ages at death (Figure 6).

Discussion

Chronological differences

The distribution and variation of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values are typically the smallest in modern Japan (Table 2, Figure 4), suggesting the dietary homogeneity in modern Japan is greater than in premodern and foraging periods. Mean $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values in modern Japan were significantly

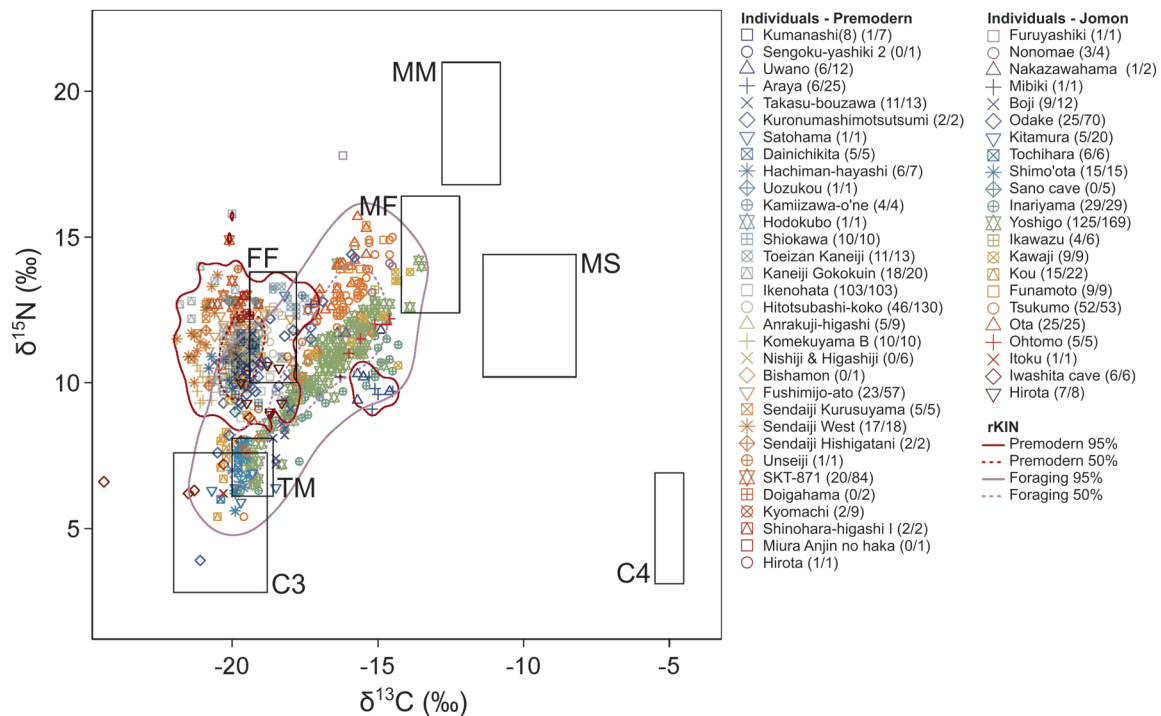


Figure 3. Distribution of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of premodern and Jomon skeletal individuals used in this study. The 50% and 95% kernel density areas of skeletal values calculated by rKIN (Eckrich et al., 2019) and the mean and 1SD square range of Japanese foods (Yoneda et al., 2004) with fixed offset are also shown. C3, C3 plants; C4, C3 plants; FF, freshwater fish; MF, marine fish; MM, marine mammals; MS, marine shellfish; TM, terrestrial mammals. Numbers in parenthesis are the accepted number and the total number of subject individuals from the site.

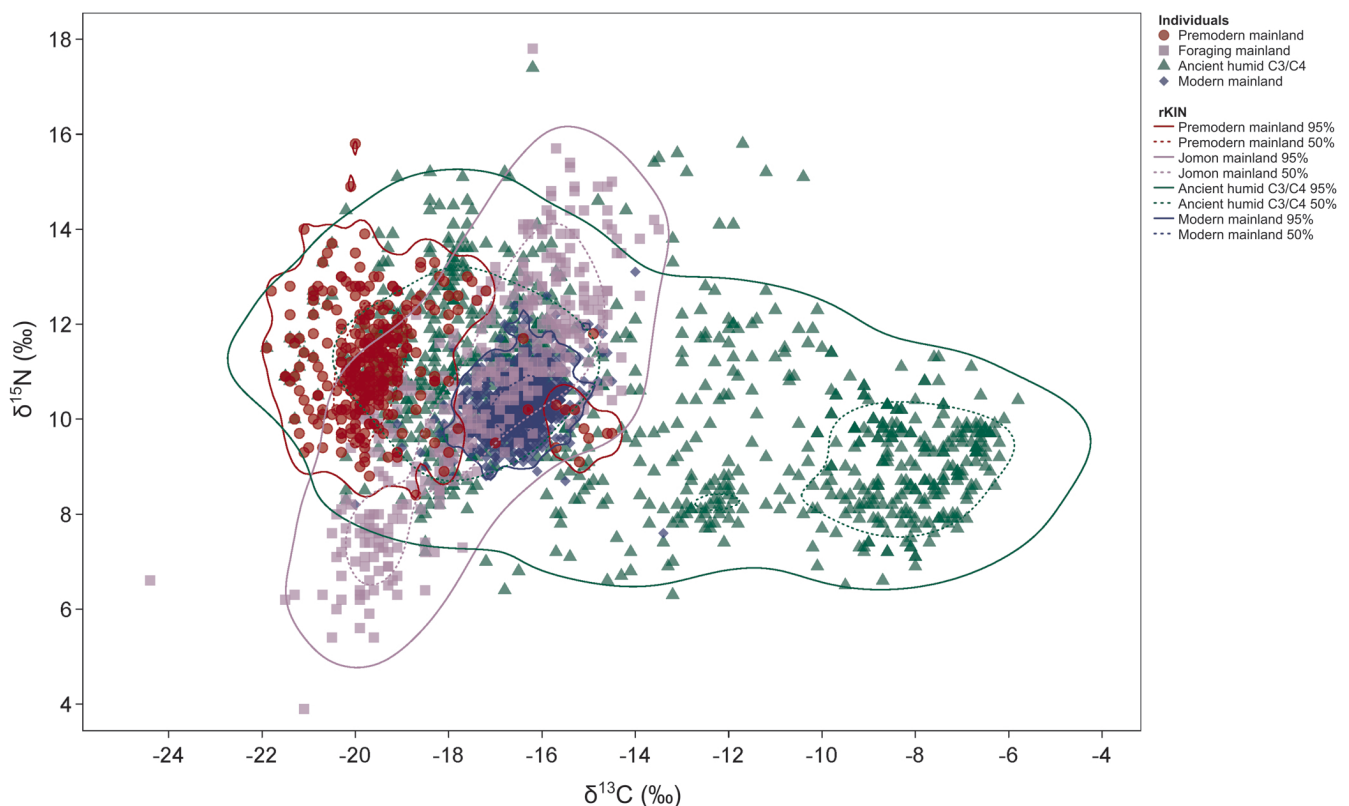


Figure 4. Distribution of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of premodern and Jomon skeletal individuals from mainland Japan, human remains from ancient humid C3/C4 vegetation regions (Bird et al., 2021), and hairs from modern Japan with fixed offset (Kusaka et al., 2016). The 50% and 95% kernel density areas of the stable isotope ratios calculated by rKIN (Eckrich et al., 2019) are also shown.

Table 3. Results of Mann–Whitney *U*-tests applied to the combination of different datasets

	$\delta^{13}\text{C}$		$\delta^{15}\text{N}$	
	<i>U</i>	<i>P</i>	<i>U</i>	<i>P</i>
Premodern–Modern	12186.0	<0.001	244687.5	<0.001
Premodern–Jomon	16952.0	0.011	62876.5	0.011
Modern–Jomon	216031.0	<0.001	153982.5	<0.001

Table 4. Areas of the 50% and 95% kernel density distributions of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values for different datasets

	50% area	95% area
Jomon	13.6	49.6
Premodern	3.4	22.9
Modern	1.1	6.5
Ancient humid C3/C4	27.5	106.0

greater and smaller than in premodern Japan (Table 3), which represents the increased contribution of C4 crops and domesticated animals fed by C4 plants and a decreased contribution of marine fish, respectively (Kusaka et al., 2016). These trends are consistent with the stable isotope

change seen in the United States driven by the “global supermarket effect” (Valenzuela et al., 2012).

Dietary homogenization had occurred even in the premodern period in mainland Japan, but to a smaller degree than that seen in modern Japan. The 95% kernel density area of the premodern period is 46% of that of Jomon period, while that of modern Japan represents only 13% (Table 4). Compared with the Jomon period, the distribution of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values was significantly smaller in the premodern mainland and contracted to the intermediate position between C3 crops and marine fish (Figure 3). Although a detailed investigation of feeding ecology during the foraging periods is beyond the scope of this study, distributions and summary statistics of each population data showed the dietary homogenization proceeded at both inter- and intra-population levels from Jomon to premodern and from premodern to modern periods (Figure 4). The dietary homogenization in premodern mainland Japan would be a result of (i) the increased access to staple protein sources (e.g. rice and fish) because of the developed food production system and commerce network; and (ii) the relatively homogenized sense of values toward diet because of the stable nationwide reign by the Tokugawa Shogunate (Ehara et al., 2009; Harada, 2009; Ishige, 2015).

Table 5. Results of Mood tests applied to the combination of different datasets

	$\delta^{13}\text{C}$		$\delta^{15}\text{N}$	
	<i>U</i>	<i>P</i>	<i>U</i>	<i>P</i>
Premodern–Modern	6.8	<0.001	13.6	<0.001
Premodern–Humid C3/C4	−19.4	<0.001	−9.1	<0.001
Premodern–Jomon	−11.6	<0.001	−10.8	<0.001
Modern–Humid C3/C4	−35.5	<0.001	−27.1	<0.001
Modern–Jomon	−23.2	<0.001	−22.9	<0.001
Jomon–Humid C3/C4	17.3	<0.001	−6.3	<0.001

Stable isotope ratios were standardized by subtracting the average.

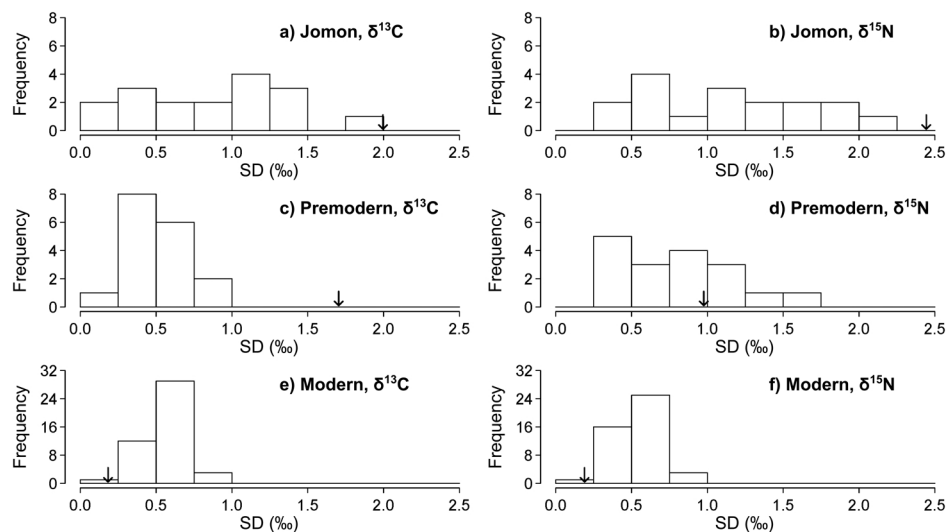


Figure 5. Distribution of 1SD of $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of skeletal individuals from each site or prefecture in (a, b) Jomon, (c, d) premodern, and (e, f) modern mainland Japan, which represent intra-site variations. The vertical arrows show 1SD of average $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values for sites/prefectures that have ≥ 3 acceptable individuals, which represent inter-site variations.

Table 6. Summary information of the premodern archaeological sites used in this study

Site	Japanese name	$\delta^{13}\text{C}$		$\delta^{15}\text{N}$		n		
		Mean	SD	Mean	SD	Accepted	Conservative	Reported
Kumanashi (8)	隈無(8)	-19.3	—	11.3	—	1	1	7
Sengoku-yashiki 2	千石屋敷 2	—	—	—	—	—	—	1
Uwano	上野	-15.5	0.6	10.5	1.0	6	6	12
Araya	荒谷	-15.2	0.6	9.8	0.4	6	6	25
Takasu-bouzawa	鷹巣坊沢	-19.6	0.4	10.7	0.8	11	4	13
Kuronumashimotsutsumi-shitagate-ato	黒沼下堤下館跡	-20.1	0.3	9.2	0.2	2	—	2
Satohama	里浜	-18.2	—	13.0	—	1	—	1
Dainichikita	大日北	-19.5	0.3	11.2	0.4	5	2	5
Hachiman-hayashi	八幡林	-20.2	0.5	10.3	0.5	6	—	7
Uozukou	魚津港	-20.1	—	12.8	—	1	1	1
Kamiizawa-o'ne	上居沢尾根	-17.8	1.0	12.0	1.7	4	—	4
Hodokubo	程久保	-17.4	—	12.5	—	1	—	1
Shiokawa	塩川	-19.1	0.9	9.4	0.4	10	10	10
Toeizan Kaneiji	東叡山寛永寺	-18.3	0.6	12.4	0.9	11	11	13
Kaneiji Gokokuin	寛永寺護国院	-20.3	0.7	12.4	1.3	18	—	20
Ikenohata-shichikencho	池之端七軒町	-19.7	0.4	10.8	0.7	103	103	103
Hitotsubashi	一橋高校	-19.4	0.5	11.0	0.8	46	46	130
Anrajuji-higashi	安楽寺東	-18.3	0.2	10.2	0.7	5	5	9
Komekuyama B	米倉山 B	-20.8	0.4	9.9	0.5	10	10	10
Nishiji and Higashiji	西地・東地	—	—	—	—	—	—	6
Bishamon	毘沙門洞窟	—	—	—	—	—	—	1
Fushimijo-ato	伏見城跡	-19.8	0.7	12.1	0.6	23	23	57
Sendaiji Kurusuyama	千提寺クルス山	-21	0.3	11.5	1.1	5	5	5
Sendaiji West	千提寺西	-21.1	0.4	11.6	1.0	17	17	18
Sendaiji Hishigatani	千提寺菱ヶ谷	-20.8	0.1	10.9	1.3	2	1	2
Unseiji	雲晴寺	-19.8	—	13.9	—	1	1	1
Sakai-kango-toshi 871	堺環濠都市 871 地点	-20	0.3	12.3	1.0	20	20	84
Doigahama	土井ヶ浜	—	—	—	—	—	—	2
Kyomachi	京町	-19.6	0.1	12.3	0.1	2	2	9
Shinohara-higashi I	篠原東 I	-19.5	0.1	12.6	0.5	2	2	2
Miura Anjin no haka	三浦按針の墓	—	—	—	—	—	—	1
Hirota	廣田	-19.5	—	10.3	—	1	—	1

References for individual sites are shown in the datasets (10.48530/isoarch.2021.006 and 10.48530/isoarch.2023.003) published in the IsoArch repository (Plomp et al., 2022; Salesse et al., 2018).

Comparison with the other datasets suggests that the stable isotope niche of the premodern mainland diet has the unique characteristics of contribution from fertilized C3 crops and/or rice influenced by denitrification in paddy fields. Individuals from several sites (e.g. Kaneiji Gokokuin, Sendaiji Kurusuyama, Sendaiji West, and Unseiji) showed lower $\delta^{13}\text{C}$ but higher $\delta^{15}\text{N}$ values (Figure 2, Table 6). Individuals from such a lower- $\delta^{13}\text{C}$ but higher- $\delta^{15}\text{N}$ niche were not seen in the Jomon period (Figure 3) and represent an extreme niche of the isotope distribution of individuals from ancient humid C3/C4 vegetation regions of the world (Figure 4). The most plausible causes of the emergence of such an isotope niche are (i) the development of organic fertilizers mainly derived from marine fish; and/or (ii) denitrification occurring in paddy fields.

Several historical documents have described the inten-

sive use of marine fish, especially sardines, and human manure as fertilizers for crops in the premodern period (Ando et al., 1982; Hanley, 1997; Ehara et al., 2009; Harada, 2009). Crops incorporate nitrogen from organic fertilizers made from marine organisms or manure that are relatively enriched in ^{15}N (Szpak, 2014) while they incorporate carbon from the atmosphere by photosynthesis. Therefore, fertilized C3 crops increased their $\delta^{15}\text{N}$, and consumption of such crops increased human $\delta^{15}\text{N}$ but not $\delta^{13}\text{C}$ values. Studies using sulfur isotopes, which is an indicator of marine contribution, also suggested the importance of marine fertilizer for crops in premodern mainland Japan (Tsutaya et al., 2016a).

Furthermore, some studies suggested that denitrification in the paddy rice field (Mariotti et al., 1988; Xing et al., 2002) is another possible cause of higher $\delta^{15}\text{N}$ values seen

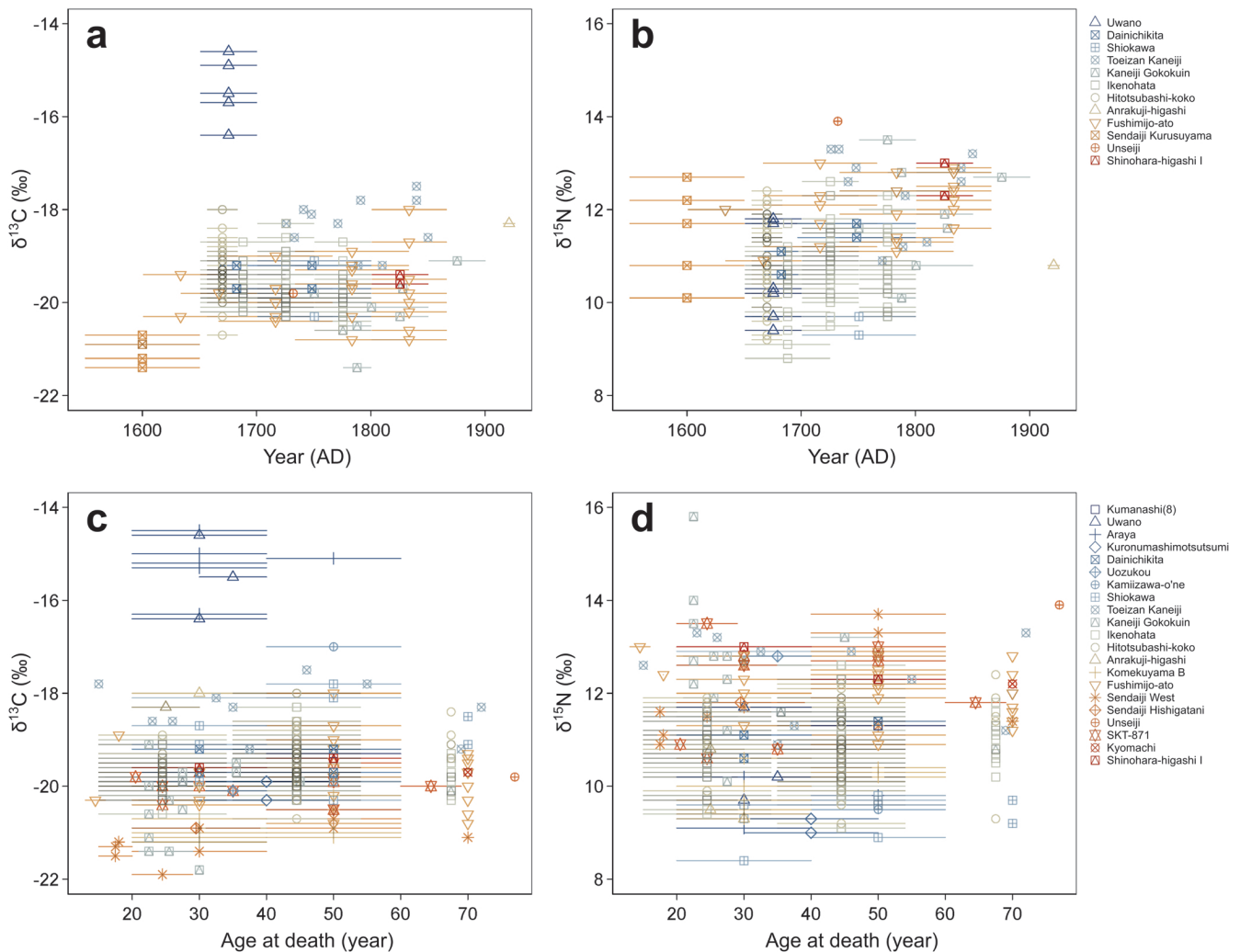


Figure 6. Relationship between the stable isotope ratios and (a, b) chronological period of the premodern individuals and (c, d) biological age at death of the premodern individuals. Error ranges are also shown with bars.

in premodern human skeletons in Japan (Tsutaya et al., 2019). In aerobic conditions at surface water depth, organic nitrogen sources (e.g. plant remains, animal tissues, and excreta) are decomposed into ammonium (NH_4^+), which is then nitrified to nitrate (NO_3^-) by bacteria. In anaerobic conditions at deeper water depths, denitrification occurs whereby nitrate is reduced by denitrifying bacteria, and the denitrified nitrogen is released into the atmosphere as gaseous N_2 . Heavy nitrogen (^{15}N) is enriched in the remaining soluble nitrogen compounds through the denitrification process, and plants growing with this nitrogen source show higher $\delta^{15}\text{N}$ values (Mariotti et al., 1988; Yoneyama et al., 1990; Xing et al., 2002). Therefore, rice affected by denitrification increased its $\delta^{15}\text{N}$, and heavy reliance on such rice products as a dietary protein source (Harada, 2009) increased human $\delta^{15}\text{N}$ but not $\delta^{13}\text{C}$ values.

Regional and individual differences

There are several individuals showing clear evidence of the consumption of C4 crops in premodern Japan. Despite their similar cultural backgrounds compared with the other

populations, it is interesting that two populations from northern Japan, Tohoku (Uwano and Araya), are separated from the 95% kernel density areas of the rest of the population and show a slight contribution of C4 crops (Figure 2). In Tohoku, the ecological setting was not ideal for paddy rice farming, and people from these sites would supplement C4 crops in their diet (Harada, 2009).

The significantly higher $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values of individuals from Toeizan Kaneiji provide evidence of dietary differences between ordinary people and the Shogun family in the highest social class in premodern Japan (Figure 2). Historical documents showed that the diet of the Shogun family was slightly rich in fish proteins compared with the diet of ordinary people (Harada, 2009). Because marine fish is enriched in ^{13}C and ^{15}N compared with terrestrial C3 foods (Yoneda et al., 2004), the isotope results reflect such a difference. However, the isotope difference seen between the Shogun family and ordinary people (1.3‰ for both $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) is relatively small in a stable isotope ecology where the $\delta^{15}\text{N}$ values of organisms increase 3–4‰ with the increase of one trophic level (Minagawa and Wada,

1984; Schoeninger and DeNiro, 1984; Lee-Thorp, 2008). Therefore, although there was a slight isotope difference, this result, as well as the lack of any observed trend in the age and period of the individuals and the small degree of sex differences (Figure 6), indicates a relatively isotopically homogeneous diet in premodern mainland people (Tsutaya et al., 2016b).

Significance and limitations

The significance of this study is the comprehensive reconstruction of the actual diet in the premodern period of mainland Japan, which supports evidence obtained in archaeology and history and reveals hidden dietary characteristics. Although a wealth of archaeological remains and historical documents provide information on the diet in the premodern period (Ehara et al., 2009; Harada, 2009), empirical information on the everyday diet of ordinary people is still lacking. This is because the amount of archaeological remains is not necessarily equal to the proportion of food sources actually eaten by an individual, and historical documents are usually biased toward people of higher social status. In this study, meta-analysis of stable isotope ratios of human skeletons yielded general patterns and regional differences in individual-level diet. As suggested by archaeology and history, stable isotope data showed that the main protein sources were C3 crops and fish in premodern Japan (Figure 3). This meta-analysis newly revealed that the established premodern food production systems created new isotope niches in mainland Japan due to the heavy reliance on fertilized C3 crops and/or paddy rice. The Shogun family was identified as having a relatively unique diet rich in marine resources. By combining archaeological/historical evidence and comprehensive stable isotope data, the ancient diet can be investigated more widely in finer resolution.

The major limitation of this study is the sampling bias in the analyzed datasets. Stable isotope data of human skeletons from premodern mainland Japan is biased toward large cities, such as Edo (i.e. present Tokyo), because modern cities in Japan are usually developed from premodern cities, where the frequency of rescue excavation campaigns is greater due to urban development. Larger premodern cities typically had a greater number of populations and larger cemeteries. Therefore, the greater chance of excavation and the larger number of inhabitants increase the number of reported skeletal individuals with stable isotope ratios in cities. The difference in the distribution of the subject sites between Jomon and premodern periods would partly reflect this fact (Figure 1). Due to the incompleteness of the stable isotope data, Central and South Ryukyus and Hokkaido were excluded from this study. Several stable isotope studies, as well as archaeological studies, showed marine mammals and fish to be the most important protein source of people living in ancient Hokkaido, and ancient people in Hokkaido had a different isotope niche compared to ancient people from the mainland (Minagawa, 2001; Naito et al., 2010; Yoneda et al., 2011; Tsutaya et al., 2014). Food sources from subtropical environments, which typically show different stable isotope ratios with temperate environments, characterize the ancient diet in Ryukyu.

Bone collagen reflects the average diet on the decadal scale and gives little information about shorter-term dietary fluctuations and detailed dietary items. To reconstruct short-term fluctuations in premodern diet, time-resolving tissues, such as hairs (Maruyama et al., 2018) and tooth dentine (Tsutaya, 2020), need to be analyzed. Consumption of different dietary items which show similar stable isotope ratios cannot be distinguished. To investigate detailed dietary items, ancient DNA and palaeoproteomic analyses of food remains in dental calculus are promising methods for accurate and comprehensive identification of the consumed food resources (e.g. Hendy et al., 2018; Sawafuji et al., 2020).

Conclusions

Carbon and nitrogen stable isotope ratios of adult human skeletons from premodern mainland Japan were compiled and compared to those reported from the Jomon period and in modern Japan. The comparison showed that the human diet was rapidly homogenized isotopically in modern Japan. The isotope diversity of human diets in mainland Japan decreased from Jomon to premodern periods. Premodern people in mainland Japan typically obtained dietary proteins from C3 crops and fish. The establishment of agriculture created new isotope niches for mainland Japan due to the heavy reliance on fertilized C3 crops and/or paddy rice which show higher $\delta^{15}\text{N}$ and lower $\delta^{13}\text{C}$ values. The Shogun family, people of the highest social class, showed unique diets that would relate to the individuals' social status or availability of foods. This study contributes to the comprehensive understanding of human dietary change through time and regional differences in the human diet.

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Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Conceptualization: T.T., R.S.; resources: N.D., C.K.; formal analysis: T.T.; funding acquisition: T.T., R.S., Y.M.; investigation: T.T., N.D., C.K.; methodology: T.T., Y.M.; project administration: T.T., R.S., Y.M.; visualization: T.T.; writing—original draft: T.T.; writing—review and editing: T.T.

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